Roads Are No Fun

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Levels of Assembly

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1 Levels of Assembly

Levels of assembly are often a matter of perspective; the goal is for the distinction to be useful for a product life cycle. For design, certain requirements only affect specific levels of assembly, which is the purpose of this document. Each level is critical to high reliability systems.

1.1 Part

A part is a single piece or substance.
examples: bracket, spring, lubricant
Parts, Materials, and Processes
Materials and processes are usually controlled by industry standards.
Parts typically require a drawing which identify relevant materials and processes.

1.2 Component

A component is a basic functional element consisting or two or more parts.
examples: ball bearing, gear set, honeycomb panel
May require a procurement specification.

1.3 Subassembly

A subassembly is a collection of components and / or other subassemblies.
examples: electric motor, cable wrap, slip ring
May require a procurement specification.

1.4 Assembly

An assembly is a complete functional module consisting of multiple subassemblies.
examples: gimbal, reaction wheel, antenna, rover moblity corner
Requires a specification

1.5 Subsystem

A subsystem encompasses all elements of a functional set.
examples: communications, propulsion, attitude control, rover mobility, power, computing
Requires a specification

1.6 System

Mission level
example: spacecraft, rover, ground vehicle,
Requires a specification

1.7 Enterprise / System of Systems

Campaign level

Consists of multiple systems

1.8 Interfaces

References

1. MIL-A-83577, 1988.